Document: Distributed Systems and Data Infrastructure

1\. Introduction to Distributed Systems

A distributed system is a collection of independent computers that communicate over a network and coordinate their activities to achieve a common objective. Unlike a traditional monolithic application, where components typically execute within a single process, distributed systems divide responsibilities across multiple machines or services. This architecture improves scalability, fault isolation, and resource utilization. However, it also introduces challenges such as network latency, partial failures, data consistency, and synchronization. Examples include cloud computing platforms, distributed databases, microservices architectures, and large-scale search engines.

2\. Database Replication

Database replication is the process of maintaining copies of the same data across multiple database servers. Replication improves availability and can distribute read traffic across several nodes. In a primary-replica architecture, the primary database handles write operations while replica databases receive copies of the changes. Synchronous replication requires replicas to acknowledge changes before a transaction is considered complete, whereas asynchronous replication allows the primary to confirm transactions without waiting for replicas. Although asynchronous replication generally provides better write performance, it introduces the possibility of temporarily stale data when replicas have not yet received recent updates.